Raffles Institution
H2 Mathematics (9758)
Solution for 2024 A-Level Paper 1

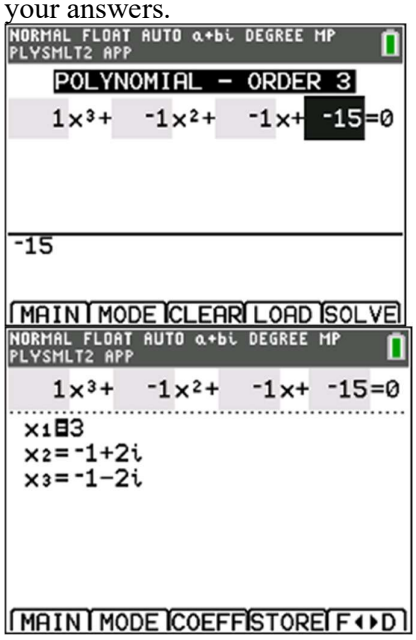
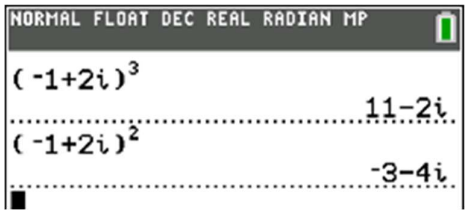
Question 1

	Suggested Solution	Remarks for Student
[4]	$y = \dfrac{x+1}{ax^2+bx+c}$ Asymptote at $x = -\dfrac{1}{2} \Rightarrow a\left(-\dfrac{1}{2}\right)^2 + b\left(-\dfrac{1}{2}\right) + c = 0$ $\Rightarrow a - 2b + 4c = 0 \quad \cdots(1)$ Graph passes through $\left(-2, -\dfrac{1}{9}\right) \Rightarrow -\dfrac{1}{9} = -\dfrac{1}{4a-2b+c}$ $\Rightarrow 4a - 2b + c = 9 \quad \cdots(2)$ $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{ax^2+bx+c-(x+1)(2ax+b)}{(ax^2+bx+c)^2}$ $= \dfrac{-ax^2-2ax+c-b}{(ax^2+bx+c)^2}$ Turning point at $\left(-2, -\dfrac{1}{9}\right) \Rightarrow -4a + 4a + c - b = 0$ $\Rightarrow -b + c = 0 \quad \cdots(3)$ Solving: $a = 2,\ b = -1$ and $c = -1$	

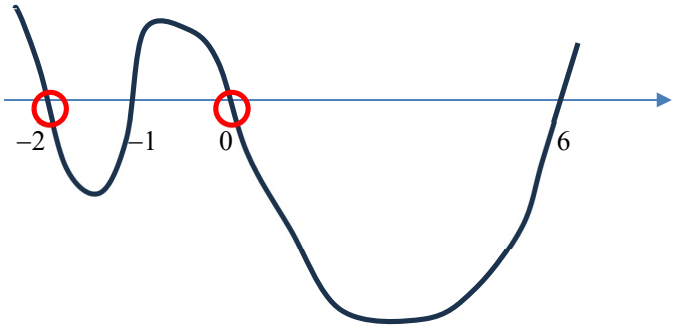
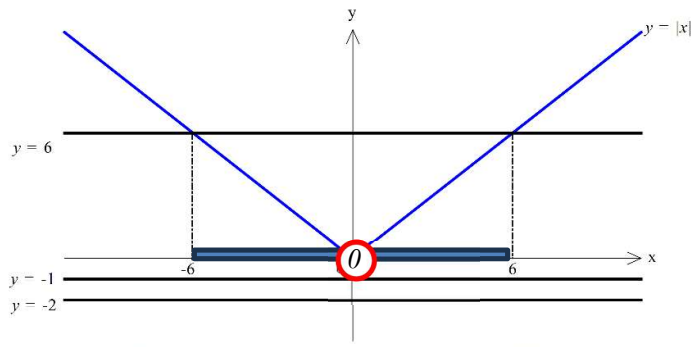
Question 2

	Suggested Solution	Remarks for Student
[3]	$\mathbf{b}.\mathbf{b} = 9 \Rightarrow \lvert\mathbf{b}\rvert^2 = 9 \Rightarrow \lvert\mathbf{b}\rvert = 3$ $\mathbf{a}$ is a unit vector $\Rightarrow \lvert\mathbf{a}\rvert = 1$ $\lvert\mathbf{a}\times\mathbf{b}\rvert = 3 \Rightarrow (1)(3)\sin\theta = 3$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$ $\Rightarrow \sin\theta = 1$ $\Rightarrow \theta = 90°$ $\therefore$ $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.	Sufficient working needs to be shown to support the answer which is given in the question.

Question 3

	Suggested Solution	Remarks for Student
(a) [1]	$a,\ b$ and c must be real numbers.	
(b) [3]	Method 1 (preferred): $z^3+az^2+bz+c \equiv (z-3)\left[z-(-1+2\mathrm{i})\right]\left[z-(-1-2\mathrm{i})\right]$ $\equiv (z-3)\left[(z+1)^2-(2\mathrm{i})^2\right]$ $\equiv (z-3)\left[z^2+2z+5\right]$ $\equiv z^3-z^2-z-15$ $\therefore a=b=-1$ and $c=-15$	You can use GC Plysmlt2 to check your answers. 
	Method 2: $w=3$ is a root $\Rightarrow 3^3+3^2a+3b+c=0 \quad \Rightarrow 9a+3b+c=-27 \quad \cdots(1)$ $w=-1+2\mathrm{i}$ is a root $\Rightarrow (-1+2\mathrm{i})^3+(-1+2\mathrm{i})^2a+(-1+2\mathrm{i})b+c=0$ $\Rightarrow 11-2\mathrm{i}+(-3-4\mathrm{i})a+(-1+2\mathrm{i})b+c=0$ Comparing real part: $-3a-b+c=-11 \quad \cdots(2)$ Comparing imaginary part: $-4a+2b=2 \quad \cdots(3)$ Solving using GC Plysmlt2: $a=b=-1$ and $c=-15$	Note that we can use GC to evaluate $(-1+2\mathrm{i})^3, (-1+2\mathrm{i})^2$ etc 

Question 4

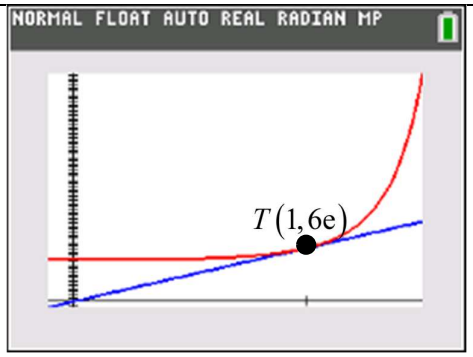
No.	Suggested Solution	Remarks for Student
(a) **[4]**	$\dfrac{4}{x+2} \geq \dfrac{x-3}{x}$ $\Rightarrow \dfrac{4x-(x+2)(x-3)}{x(x+2)} \geq 0$ $\Rightarrow \dfrac{-4x+(x+2)(x-3)}{x(x+2)} \leq 0$ $\Rightarrow x(x+2)(x^2-5x-6) \leq 0$ and $x \neq 0$ and $x \neq -2$ $\Rightarrow x(x+2)(x-6)(x+1) \leq 0$ and $x \neq 0$ and $x \neq -2$  $\therefore \quad -2 < x \leq -1$ or $0 < x \leq 6$	No calculator is allowed so key algebraic steps need to be included in your working Multiplying by "−1" to convert to "$+x^2$" term results in an easier algebraic manipulation that is less prone to careless mistakes
(b) **[2]**	To solve $\dfrac{4}{x+2} \geq \dfrac{x-3}{x}$, we replace x by $\lvert x \rvert$ in part (a): $-2 < \lvert x \rvert \leq -1$ or $0 < \lvert x \rvert \leq 6$ $\lvert x \rvert \geq 0 \quad \Rightarrow \quad -2 < \lvert x \rvert \leq -1$ has no solution To solve $0 < \lvert x \rvert \leq 6$, $0 < \lvert x \rvert$ and $\lvert x \rvert \leq 6$ $\Rightarrow x \neq 0$ and $-6 \leq x \leq 6$ $\therefore -6 \leq x < 0$ or $0 < x \leq 6$ Alternatively, we can consider the graph of $y = \lvert x \rvert$:  	

Question 5

No.	Suggested Solution	Remarks for Student
(a) [3]	Sum to infinity = 18 $\Rightarrow \frac{a}{1-r} = 18$ $\Rightarrow a = 18 - 18r \quad \cdots(1)$ 3rd term $= \frac{8}{3} \Rightarrow ar^2 = \frac{8}{3} \quad \cdots(2)$ (1) Into (2): $(18-18r)r^2 = \frac{8}{3} \Rightarrow 27(1-r)r^2 = 4$ $\therefore 27r^3 - 27r^2 + 4 = 0$ (shown)	
(b) [1]	Using GC, $r = -\frac{1}{3}$ or $r = \frac{2}{3}$ (reject since $r < 0$) From (1), $a = 24$.	
(c) [3]	$80\sum_{r=k+1}^{\infty} u_r = \sum_{r=1}^{k} u_r$ $80\left[\sum_{r=1}^{\infty} u_r - \sum_{r=1}^{k} u_r\right] = \sum_{r=1}^{k} u_r$ $\sum_{r=1}^{k} u_r = \frac{160}{9}$ Applying sum of GP and part (b), $\sum_{r=1}^{k} 24\left(-\frac{1}{3}\right)^{r-1} = \frac{24\left[1-\left(-\frac{1}{3}\right)^{k}\right]}{1-\left(-\frac{1}{3}\right)} = \frac{160}{9}$ $\left(-\frac{1}{3}\right)^{k} = \frac{1}{81} = \left(-\frac{1}{3}\right)^{4}$ $\therefore k = 4$	$\sum_{r=1}^{\infty} u_r = 18$ is given Alternative: $\left(-\frac{1}{3}\right)^{k} = \frac{1}{81}$ k must be even $\Rightarrow$ $\left(\frac{1}{3}\right)^{k} = \frac{1}{81}$ $k = \ln\frac{1}{81} \Big/ \ln\left(\frac{1}{3}\right) = 4$

Note that
$k \ln\left(-\frac{1}{3}\right) = \ln\left(\frac{1}{81}\right)$
is incorrect.

Question 6

No.	Suggested Solution	Remarks for Student
(a) **[5]**	$C: y = 2\mathrm{e}^{x^3} + a$, where $a > 0$ $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 2\left(3x^2\right)\mathrm{e}^{x^3} = 6x^2\mathrm{e}^{x^3}$ At T where $x = 1$, $y = 2\mathrm{e} + a$ Equation of tangent to C at T: $y - \left(2\mathrm{e} + a\right) = 6\mathrm{e}\left(x - 1\right)$ $y = \mathrm{e}\left(6x - 6 + 2\right) + a$ $y = \mathrm{e}\left(6x - 4\right) + a$, where $p = 6, q = -4, r = 1$	
(bi) **[1]**	Tangent passes through origin $0 = \mathrm{e}\left(-4\right) + a \Rightarrow a = 4\mathrm{e}$	
(bii) **[2]**	 Area of region is $\int_0^1 \left[\left(2\mathrm{e}^{x^3} + 4\mathrm{e}\right) - \left(\mathrm{e}\left(6x - 4\right) + 4\mathrm{e}\right)\right]\mathrm{d}x = 5.4 \text{ units}^2 \text{ (1dp)}$ or $\int_0^1 \left[\left(2\mathrm{e}^{x^3} + 4\mathrm{e}\right)\right]\mathrm{d}x - \frac{1}{2}(1)(6\mathrm{e})$	GC can be used to evaluate the area since exact answer is not required.

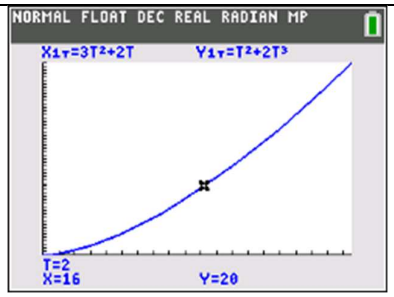
Question 7

<table>
<tr><th>No.</th><th>Suggested Solution</th><th>Remarks for Student</th></tr>
<tr><td>(a)
[2]</td><td>$\mathrm{f}(2r-1)-\mathrm{f}(2r+1)=\cos(2r-1)\theta-\cos(2r+1)\theta$
$=-2\sin\left(\frac{2r-1+2r+1}{2}\right)\theta\sin\left(\frac{2r-1-2r-1}{2}\right)\theta$
$=-2\sin(2r\theta)\sin(-\theta)$
$=2\sin\theta\sin(2r\theta)$ (shown) since $\sin(-\theta)=-\sin\theta$</td><td rowspan="2">These parts involve Factor formula and Method of Difference which are no longer in the 9758 syllabus examined from 2025 and onwards</td></tr>
<tr><td>(b)
[3]</td><td>$\sum_{r=1}^{n}\sin(2r\theta)=\frac{1}{2\sin\theta}\sum_{r=1}^{n}\mathrm{f}(2r-1)-\mathrm{f}(2r+1)$
$=\frac{1}{2\sin\theta}\begin{bmatrix}\mathrm{f}(1) & - & \mathrm{f}(3)\\ +\mathrm{f}(3) & - & \mathrm{f}(5)\\ \vdots & & \vdots\\ +\mathrm{f}(2n-1) & - & \mathrm{f}(2n+1)\end{bmatrix}$
$=\frac{1}{2\sin\theta}\left[\mathrm{f}(1)-\mathrm{f}(2n+1)\right]$
$=\frac{1}{2\sin\theta}\left[\cos\theta-\cos(2n+1)\theta\right]$ (shown)</td></tr>
<tr><td>(c)
[3]</td><td>$\sum_{r=1}^{n}\sin\left(\frac{\pi r}{6}\right)\cos\left(\frac{\pi r}{6}\right)=\frac{1}{2}\sum_{r=1}^{n}\sin 2r\left(\frac{\pi}{6}\right)$
$=\frac{1}{2}\left(\frac{\cos\frac{\pi}{6}-\cos(2n+1)\frac{\pi}{6}}{2\sin\frac{\pi}{6}}\right)$ let $\theta=\frac{\pi}{6}$ in (b)
$=\frac{1}{2}\left(\frac{\sqrt{3}}{2}-\cos(2n+1)\frac{\pi}{6}\right)$
<table><tr><td>n</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>$(2n+1)\frac{\pi}{6}$</td><td>$\frac{\pi}{2}$</td><td>$\frac{5\pi}{6}$</td><td>$\frac{7\pi}{6}$</td><td>$\frac{3\pi}{2}$</td><td>$\frac{11\pi}{6}$</td></tr><tr><td>$\cos(2n+1)\frac{\pi}{6}$</td><td>0</td><td>$-\frac{\sqrt{3}}{2}$</td><td>$-\frac{\sqrt{3}}{2}$</td><td>0</td><td>$\frac{\sqrt{3}}{2}$</td></tr></table>$\therefore$ Three possible values are $0,\frac{\sqrt{3}}{4}$ and $\frac{\sqrt{3}}{2}$.</td><td>Least value of n is 1, not 0, since least value of r is 1.</td></tr>
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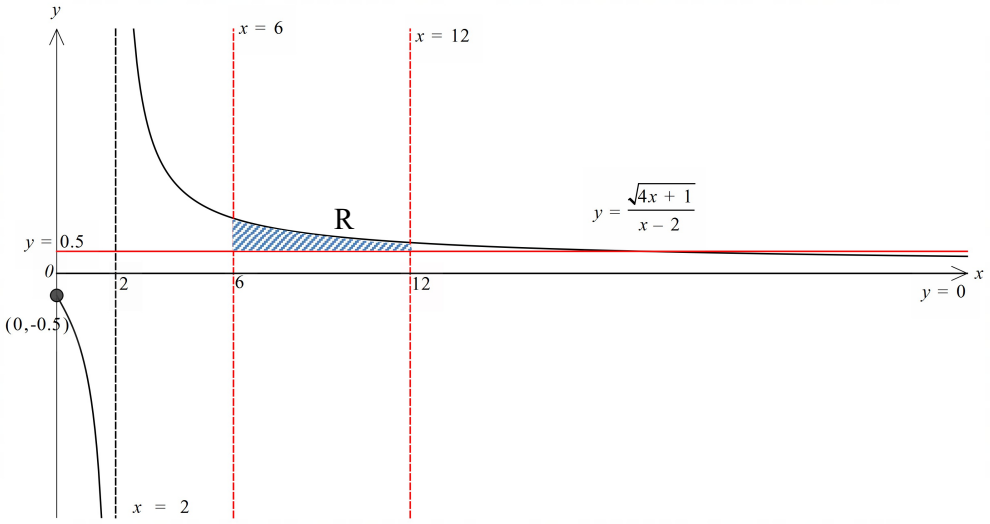
Question 8

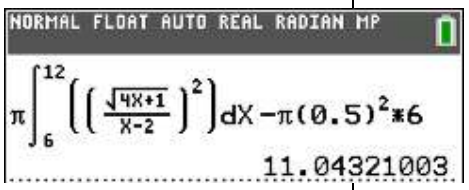
No.	Suggested Solution	Remarks for Student
(a) **[3]**	$y=\cos\left(1-\mathrm{e}^{2x}\right)$ $\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\sin\left(1-\mathrm{e}^{2x}\right)\times\left(-2\mathrm{e}^{2x}\right)$ $=2\mathrm{e}^{2x}\sin\left(1-\mathrm{e}^{2x}\right)$ $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}=2\left[2\mathrm{e}^{2x}\sin\left(1-\mathrm{e}^{2x}\right)+\mathrm{e}^{2x}\cos\left(1-\mathrm{e}^{2x}\right)\times\left(-2\mathrm{e}^{2x}\right)\right]$ $=2\left[\dfrac{\mathrm{d}y}{\mathrm{d}x}-2\mathrm{e}^{4x}y\right]$ where $k=2$ (shown) $\cdots(*)$	
(b) **[4]**	Differentiating (*) wrt x: $\dfrac{\mathrm{d}^3y}{\mathrm{d}x^3}=2\left[\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}-2\left(y\left(4\mathrm{e}^{4x}\right)+\mathrm{e}^{4x}\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)\right]$ When $x=0,\ y=\cos(1-1)=1,\ \dfrac{\mathrm{d}y}{\mathrm{d}x}=0$, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}=2[0-2]=-4$ $\dfrac{\mathrm{d}^3y}{\mathrm{d}x^3}=2[-4-2(4)]=-24$ $y=1+\dfrac{-4}{2!}x^2+\dfrac{-24}{3!}x^3+\cdots$ $=1-2x^2-4x^3+\cdots$ 1st three non-zero terms of Maclaurin expansion of $\cos\left(1-\mathrm{e}^{2x}\right)$ are $1-2x^2-4x^3$ respectively.	
(c) **[2]**	$\dfrac{1}{\sqrt{a+bx^2}}=\left(a+bx^2\right)^{-\frac{1}{2}}$ $=a^{-\frac{1}{2}}\left(1+\dfrac{b}{a}x^2\right)^{-\frac{1}{2}}$ $=a^{-\frac{1}{2}}\left[1+\left(-\dfrac{1}{2}\right)\left(\dfrac{b}{a}x^2\right)+\cdots\right]$ $=a^{-\frac{1}{2}}\left[1+-\dfrac{b}{2a}x^2+\cdots\right]$ $=a^{-\frac{1}{2}}-\dfrac{1}{2}ba^{-\frac{3}{2}}x^2+\cdots$ Comparing the two series: $a^{-\frac{1}{2}}=1\Rightarrow a=1$ $-\dfrac{1}{2}ba^{-\frac{3}{2}}=-2\Rightarrow b=4$	Factorise "a" from $\left(a+bx^2\right)^{-\frac{1}{2}}$ so that we can apply the standard series expansion of $(1+x)^n$ found in the mathematical formulae sheet on $\left(1+\dfrac{b}{a}x^2\right)^{-\frac{1}{2}}$.

Question 9

No.	Suggested Solution	Remarks for Student
(a)	$x = 3t^2 + 2t, \quad y = t^2 + 2t^3$ $\dfrac{dx}{dt} = 6t + 2, \quad \dfrac{dy}{dt} = 2t + 6t^2$ $\therefore \dfrac{dy}{dx} = \dfrac{2t + 6t^2}{6t + 2} = \dfrac{2t(1+3t)}{2(1+3t)} = t,$ where $k = 1$	
(b)	Tangent to C at point P makes an angle of 60° with the x axis $\Rightarrow$ gradient of C at $P = \tan 60^\circ = \sqrt{3}$ $\Rightarrow t = \sqrt{3}$ at P $\Rightarrow x = 3\left(\sqrt{3}\right)^2 + 2\left(\sqrt{3}\right), \quad y = \left(\sqrt{3}\right)^2 + 2\left(\sqrt{3}\right)^3$ $= 9 + 2\sqrt{3}, \qquad = 3 + 6\sqrt{3}$ Exact coordinates of P are $\left(9 + 2\sqrt{3}, 3 + 6\sqrt{3}\right)$	Exact coordinates are required
(c)	 When $x = 16, \ 16 = 3t^2 + 2t \Rightarrow t = -\dfrac{8}{3}$ (reject since $t \geq 0$) or $t = 2$ When $x = 0, \ 0 = t(3t + 2) \Rightarrow t = -\dfrac{2}{3}$ (reject since $t \geq 0$) or $t = 0$ Exact area of required region is $\int_0^{16} y \, dx$ $= \int_0^2 \left(t^2 + 2t^3\right)(6t + 2) \, dt$ $= \int_0^2 \left(12t^4 + 10t^3 + 2t^2\right) dt$ $= \left[\dfrac{12}{5}t^5 + \dfrac{5}{2}t^4 + \dfrac{2}{3}t^3\right]_0^2$ $= \dfrac{1832}{15}$ units^2 Alternatively, Exact area of required region is $16 \times 20 - \int_0^{20} x \, dy$ $= 16 \times 20 - \int_0^2 \left(3t^2 + 2t\right)\left(2t + 6t^2\right) dt$	Graph is sketched for reference GC can be used to verify the value of $\int_0^2 \left(t^2 + 2t^3\right)(6t + 2) \, dt$, although do note that exact value for area is required.

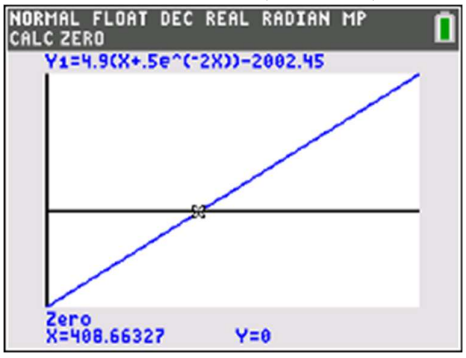
Question 10

No.	Suggested Solution	Remarks for Student
(a)	$u=\sqrt{4x+1}\Rightarrow u^2=4x+1\Rightarrow x=\dfrac{u^2-1}{4}$ and $\dfrac{\mathrm{d}x}{\mathrm{d}u}=\dfrac{2u}{4}=\dfrac{u}{2}$ $\displaystyle\int\frac{\sqrt{4x+1}}{x-2}\mathrm{d}x=\int\frac{u}{\left(\frac{u^2-1}{4}-2\right)}\left(\frac{u}{2}\right)\mathrm{d}u$ $\displaystyle=\int\frac{2u^2}{u^2-1-8}\mathrm{d}u$ $\displaystyle=\int\frac{2u^2}{u^2-9}\mathrm{d}u$ (shown)	
(bi)		
(bii))	When $x=6,\ u=5$; When $x=12,\ u=7$ Required area of R $\displaystyle=\int_6^{12}\frac{\sqrt{4x+1}}{x-2}\mathrm{d}x-(6\times0.5)$ or $\displaystyle\int_6^{12}\frac{\sqrt{4x+1}}{x-2}-0.5\ \mathrm{d}x$ $\displaystyle=\int_5^7\frac{2u^2}{u^2-9}\mathrm{d}u-3$ $\displaystyle=\int_5^7 2+\frac{18}{u^2-9}\mathrm{d}u-3$ $\displaystyle=\left[2u+\frac{18}{2\times3}\ln\left\lvert\frac{u-3}{u+3}\right\rvert\right]_5^7-3$ $\displaystyle=2(7)+3\ln\left(\frac{4}{10}\right)-2(5)-3\ln\left(\frac{2}{8}\right)-3$ $\displaystyle=1+3\ln\left(\frac{8}{5}\right)$ units^2	Exact answer is required

	Therefore, $a=1,\ b=3$ and $c=\frac{8}{5}$	
(biii)	Required volume $=\pi\int_{6}^{12}\left(\frac{\sqrt{4x+1}}{x-2}\right)^2 \mathrm{d}x-\pi(0.5)^2\times 6$ $=11.04$ units^3 (2 d.p.) 	

Question 11

No.	Suggested Solution	Remarks for Student
(ai) **[1]**	$\frac{\mathrm{d}^2x}{\mathrm{d}t^2}+\alpha\frac{\mathrm{d}x}{\mathrm{d}t}=9.8 \quad \cdots(1)$ Given $v=\frac{\mathrm{d}x}{\mathrm{d}t}$, differentiate w.r.t t, $\frac{\mathrm{d}v}{\mathrm{d}t}=\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$ Substitute into (1): $\frac{\mathrm{d}v}{\mathrm{d}t}+\alpha v=9.8$	
(aii) **[1]**	When $v=2.4$, $\frac{\mathrm{d}v}{\mathrm{d}t}=5$ $5+2.4\alpha=9.8$ $\Rightarrow \alpha=2$	
(b) **[5]**	$\frac{\mathrm{d}v}{\mathrm{d}t}+2v=9.8$ $\frac{\mathrm{d}v}{\mathrm{d}t}=9.8-2v$ $\frac{1}{9.8-2v}\frac{\mathrm{d}v}{\mathrm{d}t}=1$ Integrate w.r.t t, $\int\frac{1}{9.8-2v}\,\mathrm{d}v=\int 1\,\mathrm{d}t$ $-\frac{1}{2}\ln\lvert 4.9-v\rvert=t+c,$ where $c\in\mathbb{R} \quad \cdots(*)$ $\ln\lvert 4.9-v\rvert=-2t-2c$ $\lvert 4.9-v\rvert=\mathrm{e}^{-2t-2c}$ $4.9-v=\pm\mathrm{e}^{-2t}\mathrm{e}^{-2c}$ $4.9-v=c_1\mathrm{e}^{-2t},$ where $c_1=\pm\mathrm{e}^{-2c}$ $v=4.9-c_1\mathrm{e}^{-2t}$ When $t=0$, $v=0$. $0=4.9-c_1\Rightarrow c_1=4.9$ $\therefore v=4.9\left(1-\mathrm{e}^{-2t}\right)$ (shown), where $A=4.9$ and $B=-2$	Working involving removal of modulus needs to be shown clearly. Alternatively, since $v=0$ when $t=0 \Rightarrow 4.9-v>0$ (*) can be rewritten as $-\frac{1}{2}\ln(4.9-v)=t+c$

(c) **[5]**	$v = \dfrac{dx}{dt}$ $\Rightarrow \dfrac{dx}{dt} = 4.9\left(1 - e^{-2t}\right)$, Integrate w.r.t t, $x = 4.9\left(t + \dfrac{1}{2}e^{-2t}\right) + d$, where $d \in \mathbb{R}$ When $t = 0$, $x = 0$. $0 = 4.9\left(\dfrac{1}{2}\right) + d \Rightarrow d = -2.45$ $\therefore x = 4.9\left(t + \dfrac{1}{2}e^{-2t}\right) - 2.45$ When $x = 2000$, $2000 = 4.9\left(t + \dfrac{1}{2}e^{-2t}\right) - 2.45$ $4.9\left(t + \dfrac{1}{2}e^{-2t}\right) - 2002.45 = 0$  Solving, $t = 409$s (3 s.f) Time taken for parachutist to reach the ground is 409s.	

Question 12

No.	Suggested Solution	Remarks for Student
(ai)	Let point $A(0,0,0.1)$ represent position of air traffic controllers and l_1 represent path of the aircraft. $l_1 : \mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ Let B be foot of perpendicular from A to l_1. $\overrightarrow{OB} = \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ for some $\lambda \in \mathbb{R}$ $\overrightarrow{AB} \perp l_1 \Rightarrow \left[\begin{pmatrix} 4 \\ 3 \\ 0.9 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \right] . \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = 0$ $\Rightarrow 2.8 + 6\lambda = 0 \Rightarrow \lambda = -\frac{7}{15}$ $\overrightarrow{OB} = \begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix} - \frac{7}{15} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \frac{1}{15} \begin{pmatrix} 53 \\ 52 \\ 1 \end{pmatrix}$ Coordinates of aircraft when it is closest to air traffic controllers are $\left(\frac{53}{15}, \frac{52}{15}, \frac{1}{15} \right)$	Since the ground is modelled as the *x*-*y* plane and the base of the control tower is at (0,0,0), the tower is along the *z*-axis. Qn requires us to find coordinates, not position vector.
(aii)	$AB = \left\lvert \overrightarrow{AB} \right\rvert = \left\lvert \begin{pmatrix} 4 \\ 3 \\ 0.9 \end{pmatrix} - \frac{7}{15} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \right\rvert = \left\lvert \frac{1}{15} \begin{pmatrix} 53 \\ 52 \\ -0.5 \end{pmatrix} \right\rvert \approx 4.95 > 4$ Hence, the controllers will not see the aircraft when it is closest to them.	
(b)	Let l_2 represent the path of the drone, with equation $l_2 : \mathbf{r} = \begin{pmatrix} 1.5 \\ -1 \\ k \end{pmatrix} + \mu \begin{pmatrix} 1.5 \\ 1 \\ -1 \end{pmatrix}, \mu \in \mathbb{R}$ Observe that $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1.5 \\ 1 \\ -1 \end{pmatrix}$ are not parallel. Consider $\begin{pmatrix} 4 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1.5 \\ -1 \\ k \end{pmatrix} + \mu \begin{pmatrix} 1.5 \\ 1 \\ -1 \end{pmatrix}$	

	$\lambda - 1.5\mu = -2.5 \quad \cdots(1)$ Equivalent to $\lambda + \mu = 4 \quad \cdots(2)$ $2\lambda + \mu = k - 1 \quad \cdots(3)$ Solving (1) and (2), λ=1.4 and $\mu = 2.6$ Substituting this into (3): $k = 6.4$ For l_1 and l_2 not to intersect, (1), (2) and (3) do not have solutions, hence $k \neq 6.4$ and $k > 0$.	Note that k is a positive constant.
(c)	Acute angle between the two paths is $\cos^{-1}\dfrac{\left\lvert\begin{pmatrix}1.5\\1\\-1\end{pmatrix}.\begin{pmatrix}1\\-1\\2\end{pmatrix}\right\rvert}{\left\lvert\begin{pmatrix}1.5\\1\\-1\end{pmatrix}\right\rvert\left\lvert\begin{pmatrix}1\\-1\\2\end{pmatrix}\right\rvert}$ $= \cos^{-1}\dfrac{1.5}{\sqrt{4.25}\sqrt{6}}$ $= 72.7^\circ$ (1dp)	
(d)(i)	Distance between aircraft and drone when they are at $C(4,3,1)$ and $D(4.2,0.8,0.2)$ respectively is $\left\lvert\begin{pmatrix}0.2\\-2.2\\-0.8\end{pmatrix}\right\rvert = \sqrt{5.52} \approx 2.35$ km	As this is a real world context problem, we are required to evaluate the answer (avoid leaving it in surd form)
(d)(ii)	$\overrightarrow{CD} = \begin{pmatrix}0.2\\-2.2\\-0.8\end{pmatrix}$ Since k =2, l_1 and l_2 are skew lines (do not intersect as deduced from part (b) and they are also not parallel). For CD to be the shortest distance between the two lines, $\overrightarrow{CD}$ has to be perpendicular to both lines. $\begin{pmatrix}0.2\\-2.2\\-0.8\end{pmatrix}.\begin{pmatrix}1\\-1\\2\end{pmatrix} = 0.8 \neq 0$ $\begin{pmatrix}0.2\\-2.2\\-0.8\end{pmatrix}.\begin{pmatrix}1.5\\1\\-1\end{pmatrix} = -1.1 \neq 0$ This implies $\overrightarrow{CD}$ is not perpendicular to both lines. Hence, CD does not represent the shortest distance between the two lines.	